TECHNICAL DESIGN DOCUMENT
Township Operations Platform — Phase 5 Production Release

Pune Gated Township (2,400 Apartments · 28 Towers · 180 Facility Staff)

Target System:	Pune Gated Township (2,400 Flats, 28 Towers)
Maintenance Tariff:	■6.50 per sq. ft. per month (■7,475/mo for 1,150 sq. ft.)
Security & Roles:	Strict 2-Role RBAC (RESIDENT, HELPDESK) + PostgreSQL Force RLS
Reconciliation:	UPI VPA Normalizer + Fuzzy Matching (>= 90%) + Dual Verification
Cloud Platform:	AWS ap-south-1 (Mumbai) / ECS Fargate + RDS PostgreSQL 16 Multi-AZ

1. System Overview & Problem Statement

The platform coordinates township management across 2,400 apartments in Pune, Maharashtra. It features automated maintenance billing, multi-channel payment reconciliation (UPI/NEFT/Cheque), resident registry management, SLA-driven complaint ticketing, and committee executive reporting. All calculations strictly utilize integer paise arithmetic to guarantee financial integrity.

2. Identity, RBAC & Multi-Tenant Security

The platform maintains strictly two application roles: RESIDENT and HELPDESK. Fine-grained permissions (COMPLAINT_MANAGE, BANK_IMPORT, RECONCILIATION_REVIEW, BILLING_MANAGE, COMMITTEE_REPORT_READ) govern operational capabilities. Multi-tenant isolation is enforced at the database engine level via PostgreSQL FORCE ROW LEVEL SECURITY across all five financial tables.

3. Core Subsystem Architectural Specifications

3.1 Resident Registry & Flat Allotments

Tracks Owner, Tenant, and Family Member memberships. Tenant leases enforce date validation (valid_until >= valid_from). Expired tenant leases automatically forfeit flat data access. Mandatory E.164 phone normalization is enforced.

3.2 Billing Engine & Append-Only Financial Ledger

Invoices snapshot unit rates at generation. Balances are derived dynamically from an append-only double-entry financial ledger. The late-fee engine supports Flat Fee, Monthly Percentage, and Daily Simple Interest in basis points.

3.3 UPI-Aware Automated Bank Reconciliation

Bank statement imports undergo concurrent SHA-256 deduplication. An intelligent narration normalizer extracts UPI VPAs and fuzzy matches apartment identifiers. Matches with >= 90% confidence are promoted. A maker-checker verification modal enforces separation of duties prior to financial ledger settlement.

3.4 Service Desk & Persistent Safety Alerts

Complaints are auto-assigned via round-robin workload leveling. Emergency tickets feature a persistent duty-manager queue with a 15-minute response SLA. Staff can correct miscategorized tickets and link duplicates while preserving audit history.

3.5 Committee Reporting & PWA Resident Experience

Live SQL aggregates calculate arrears across 0-30, 31-60, 61-90, and 90+ day buckets. CSV exports include formula injection mitigation (CWE-1236). The resident portal is delivered as an installable, offline-capable Progressive Web Application (PWA).

4. Deployment & Infrastructure Architecture

AWS CDK v2 provisions Amazon VPC in Mumbai (ap-south-1), RDS PostgreSQL 16 Multi-AZ, ECS Fargate FastAPI container tasks, and CloudFront S3 static hosting. CI/CD pipelines enforce Ruff, Mypy, Pytest, Vitest, Gitleaks, and CodeQL SAST.